MDP: Markov Decision Process

① "Actions are indeterminate"

"Action" in "Search": state A $\xrightarrow{\text{Action } a}$ state B "Determinate"

Action in "MDP": state A $\xrightarrow{\text{Action } a}$ { state B $p = 0.2$; state C $p = 0.4$; state D $p = 0.4$ } "Indeterminate"

② "Cost function depends the starting and ending states of action"

MDP consists of

① a set of states * : what differs from "Search"

② a starting state

③* a set of actions modeled by a $\boxed{\text{global probabilistic transition relation}}$

④* Reward fuction $T(s, a, s')$: the probability of $s \xrightarrow{a} s'$

Reward(s, a, s'): the rewarding for $s \xrightarrow{a} s'$

⑤ Goal test: End(s)

⑥ A "discount factor": γ, $0 \le \gamma \le 1$. "How much we value the future"

Transition probabilities $T(s, a, s')$

1. if $T(s, a, s') \neq 0$ then s' is a possible successor of s
2. $\sum_{s'} T(s, a, s') = 1$

Policy: A "mapping" from "state" to "action".

$\pi: S \to A$, $\pi(s_1) = a_1$, $\pi(s_2) = a_2$ ...

How to compare Policys: "$\pi(n_1) = m_1$ (?) $\pi(n_1) = m_2$"

Expected utilities $V_\pi(s)$

How to compute V_π ?

1. By recursion.

$$V_\pi(s) = \begin{cases} 0, & \text{if End}(s) \\ \sum_{s'} T(s, \pi, s')\left[\text{Reward}(s, \pi, s') + \gamma V_\pi(s')\right], & \text{otherwise} \end{cases}$$

Q-value: "$Q_\pi(s, a) = \sum_{s'} T(s, a, s')\left[\text{Reward}(s, a, s') + \gamma V_\pi(s')\right]$"